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Power System Stability Analysis in the Presence of Line Outages using Graph Neural Networks

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Project book

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Abstract

Reliable operation of power grids depends on early detection of line outages, as undiagnosed faults can cascade into large-scale blackouts. This challenge is amplified by the growing integration of renewable energy sources, whose variability and distributed nature increase the system's complexity and vulnerability. Traditional identification methods often rely on detailed system models and state estimation, which become increasingly difficult to maintain as networks grow in size and complexity. To address this challenge, we explore a data-driven approach that leverages high-frequency phasor measurements collected across the grid. By modeling the power system as a graph, with buses as nodes and transmission lines as edges, we capture the inherent structure of the network and the spatiotemporal dynamics of outage events. Building on the architectures proposed by He and Cheng (2021), we implement and compare two graph convolutional network (GCN) variants — one operating in the spatial domain through graph shift operators, and another in the spectral domain via the graph Fourier transform. Using the IEEE 39-bus test system as a case study, we show that these models can accurately classify both single and multiple simultaneous line outages, providing a scalable framework for addressing the N - k contingency problem without requiring explicit system parameter knowledge.

Introduction

Detecting failures in power systems is crucial for maintaining grid stability and avoiding large-scale blackouts. While failures can take many forms - such as equipment malfunctions, voltage instability, or generator trips - this project focuses specifically on *line outages*, where one or more transmission lines are disconnected from service. These kinds of failures are common and can significantly disrupt the flow of power across the network.

Traditional outage detection methods rely on detailed physical models of the grid and exhaustive simulations. These approaches become increasingly impractical as the number of possible failure combinations grows, especially in the case of multiple simultaneous outages, known as the *N - k contingency problem*. In this setting, the system operator typically hypothesizes each possible outage by removing the corresponding line from the network model, runs a power flow or state estimation under this assumption, and compares the simulated results to the actual phasor measurements. This leads to a combinatorial explosion in the number of simulations required.

A key advantage of the GCN-based models we explore is their inherent scalability to this problem. Once trained, they can identify and classify multiple simultaneous line outages without changes to the network architecture or added computational cost. This makes them particularly attractive for real-time monitoring and decision support in modern, complex power systems.

Power systems can be naturally modeled as graphs, where buses are nodes and transmission lines are edges. Measurements collected by Phasor Measurement Units (PMUs) at each bus form time-series signals on this graph. This structure

allows the application of graph convolutional networks (GCNs), which are designed to capture both spatial and temporal patterns in graph-supported data.

Prior work by Routtenberg et al. (“Detection of False Data Injection Attacks in Smart Grids based on Graph Signal Processing” [5]) assumed that outage indicators from protection controllers are trustworthy while PMU data may be compromised. Our work takes the opposite stance: we rely on PMU measurements as the primary data source, without requiring direct outage signals.

In this project, we implement and evaluate two GCN-based models for identifying power line outages, following the architecture proposed by He and Cheng (2021). One model applies graph convolutions in the spatial domain using graph shift operators, while the other operates in the spectral domain using the graph Fourier transform (GFT). Both models are designed to classify graph signals associated with different outage scenarios, without relying on explicit system parameters.

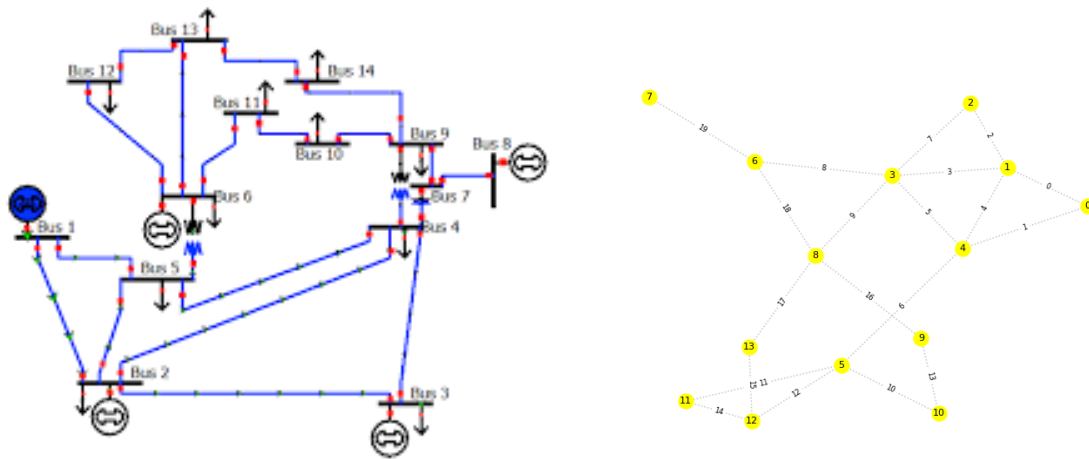


Figure 1: IEEE 14 system as a graph

GCNs and GFT

Graph Convolutional Networks (GCNs) are a type of Graph Neural Network specifically designed to extend the concept of convolution networks to graph-structured data. The key idea is to define a way to combine (or “filter”) information from a node and its neighbors, based on the topology of the graph.

In a regular grid like an image, convolution is performed using a kernel that slides over fixed local neighborhoods. However, in a graph, the notion of a neighborhood is irregular and defined by connectivity between nodes. GCNs address this by using matrix operations that depend on the graph structure.

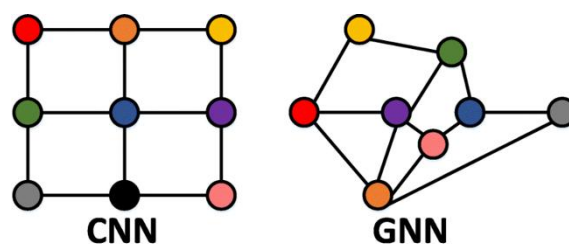


Figure 2: GCNs (or GNNs) operate similarly to CNNs on graph-structured data

Spatial Domain Convolution

In the spatial GCN architecture, filtering is done in the vertex (node) domain using a graph shift operator. In our case the operator is based on the normalized adjacency matrix, denoted as:

$$\tilde{A} = D^{-\frac{1}{2}}AD^{-\frac{1}{2}}$$

Here, A is the adjacency matrix of the graph, and D is a diagonal matrix containing the degrees of the nodes. This normalization helps ensure numerical stability.

Given a graph signal $x \in \mathbb{R}^N$ ($N = \text{\#nodes}$), where each element represents the signal at one node, we can define a simple graph convolution operation as:

$$y = w_0x + w_1\tilde{A}x$$

This equation means that the new signal y at each node is a weighted combination of the original value at that node and the values at its direct neighbors (1-hop neighbors). w_0 and w_1 are learnable weights. We can generalize this to a higher-order filter (i.e., include information from neighbors that are 2-hops, 3-hops, etc.) using a polynomial of $\tilde{A}$:

$$y = \sum_{i=0}^H w_i \tilde{A}^i x$$

This is known as a vertex-domain filter, where H is the number of hops considered (the filter's order). The signal is repeatedly multiplied by $\tilde{A}$, propagating information across the graph structure.

When using multiple features as input, the input becomes a matrix $X \in \mathbb{R}^{N \times K}$, where K is the number of features per node. The output of the convolution layer is:

$$Y = \sum_{i=0}^H \tilde{A}^i X W_i$$

where each $W_i \in \mathbb{R}^{K \times G}$ is a trainable weight matrix, and G is the number of output features per node. One can think of the input and output features at each node as analogous to the input and output channels in a conventional CNN.

GFT

Similarly to how the Discrete Fourier Transform (DFT) can be computed via matrix multiplication, the Graph Fourier Transform (GFT) can also be applied to a graph signal through matrix operations. Recall that in the classical setting, the DFT matrix diagonalizes the cyclic convolution matrix, allowing convolution in the time domain to be expressed as pointwise multiplication in the frequency domain:

$$h * x = Hx = V^T \Lambda Vx$$

Where H is the convolution matrix and V is the DFT matrix. This is known as the convolution theorem.

A similar principle applies to graph signals. For a graph with a symmetric normalized adjacency matrix $\tilde{A}$, we can perform an eigen decomposition:

$$\tilde{A} = V\Lambda V^T$$

Here, V contains the eigenvectors of $\tilde{A}$, and Λ is a diagonal matrix of its eigenvalues. Since $\tilde{A}$ plays a role analogous to a convolution operator on graph signals, its eigen decomposition forms the basis for the graph Fourier transform. The GFT of a graph signal x is then given by:

$$\hat{x} = V^T x$$

That is, the GFT matrix is V^T , and it transforms the signal from the vertex (spatial) domain to the graph frequency domain.

Spectral Domain Convolution

Convolution in the spectral domain is defined as:

$$\hat{y} = w(\Lambda)\hat{x}$$

where $w(\Lambda) = \sum_{i=0}^H w_i \Lambda^i$ is the spectral filter, a polynomial in the eigenvalues. In the Spectral GCN architecture, the model operates entirely in the frequency domain and uses $\hat{y}$ directly as input to subsequent layers such as classifiers. This allows the network to learn and make predictions based purely on spectral components, without transforming the signal back to the spatial domain.

Alternatively, applying the inverse GFT to $\hat{y}$:

$$y = V\hat{y} = Vw(\Lambda)V^T x = \sum_{i=0}^H w_i \tilde{A}^i x$$

returns the convolved signal to the vertex domain (equivalent to performing the spatial domain convolution).

For K input features and G output features, the g -th output in the spectral domain becomes:

$$\hat{y}^{(g)} = \sum_{k=1}^K w^{(k,g)}(\Lambda) \hat{x}^{(k)}, \quad g = 1, \dots, G$$

i.e. learning a kernel for each features combination.

Analyzing graph signals in the spectral domain offers a natural and powerful way to capture the global structure of the graph. By decomposing the signal into its graph frequency components using the Graph Fourier Transform (GFT), it becomes possible to distinguish between smooth, low-frequency patterns and sharp, high-frequency variations. In the context of power systems, line outages typically introduce localized disturbances that correspond to higher-frequency modes on the graph. Operating in the spectral domain allows the model to detect and emphasize these frequency-specific features, which may be harder to isolate when working

directly in the vertex domain. This frequency-based perspective provides a principled framework for learning filters that are sensitive to important structural disruptions in the graph signal.

Why Use GCNs for Power Line Outage Detection?

GCNs are well suited for this task because power systems are naturally represented as graphs: nodes correspond to buses, and edges to transmission lines. Measurement data like voltage angle time series can be treated as graph signals. GCNs can then be trained to detect outage patterns by learning from these signals across time and space (the graph).

Moreover, the architecture used in this project supports **multi-line outage detection** (N-k contingency) directly. Since the model outputs the predicted status for each line independently, it can detect multiple outages without modifying the network structure or adding computational complexity.

Architecture

The paper proposes two parallel but structurally similar GCN pipelines - one operating in the vertex (spatial) domain and the other in the graph-frequency (spectral) domain. Each branch applies an H-degree polynomial graph filter, followed by a non-linearity and a lightweight classifier that predicts the status of every transmission line simultaneously.

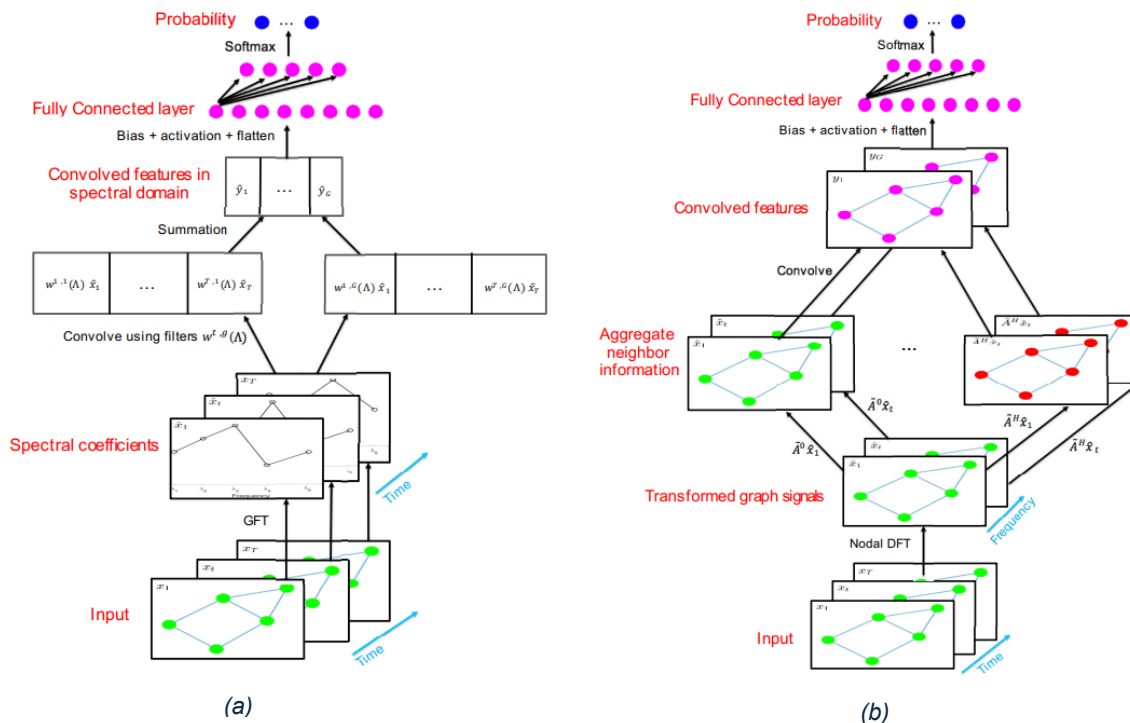


Figure 3: He & Cheng spectral (a) and spatial (b) architectures.

Data

To train and evaluate the GCN architectures, we generated synthetic data using the IEEE 39-bus power system as the test case. At each bus, the relevant signal is the time series of voltage phase angles, which are influenced by changes in the grid topology, such as line outages. All simulations and power flow calculations were conducted using the open-source Python package pandapower. 

The dataset was created by simulating various line outage scenarios. Each simulation consists of a short time window of steady-state measurements, during which one or more transmission lines are removed from service. These outages create detectable disturbances in the angle measurements across the network, which serve as input to the model.

Synthetic-Data Generation

Training examples were produced with a straightforward four-step routine:

1. **Baseline solution.**

The IEEE-39 test grid was solved once in its normal, fault-free state to capture reference voltage-angle values at all buses.

2. **Contingency catalogue.**

Three types of scenarios were enumerated:

- the no-fault case,
- every single transmission-line outage ($N-1$), and
- every unordered pair of simultaneous outages ($N-2$).

Each scenario was replicated 40 times so that different noise patterns could be applied later.

3. **40-sample window with a fix fault step.**

For every replica, 40 independent steady-state power-flow calculations were run. Because each sample corresponds to a separate steady-state solution, the sample index serves only as a proxy for “time,” but no dynamic simulation is performed. An outage is introduced at step 20.

4. **Load-noise injection.**

In every sample the active and reactive loads at all buses were randomly perturbed by up to $\pm 3\%$.

After solving the power flow, the vector of bus voltage angles was stored as the input signal, together with a binary label indicating which line(s) were out of service.

The resulting 40-point angle traces – 40 replicas of every $N-0$, $N-1$, and $N-2$ contingency - combine clear outage signatures with mild, realistic variability, providing suitable training data for the graph-convolutional models.

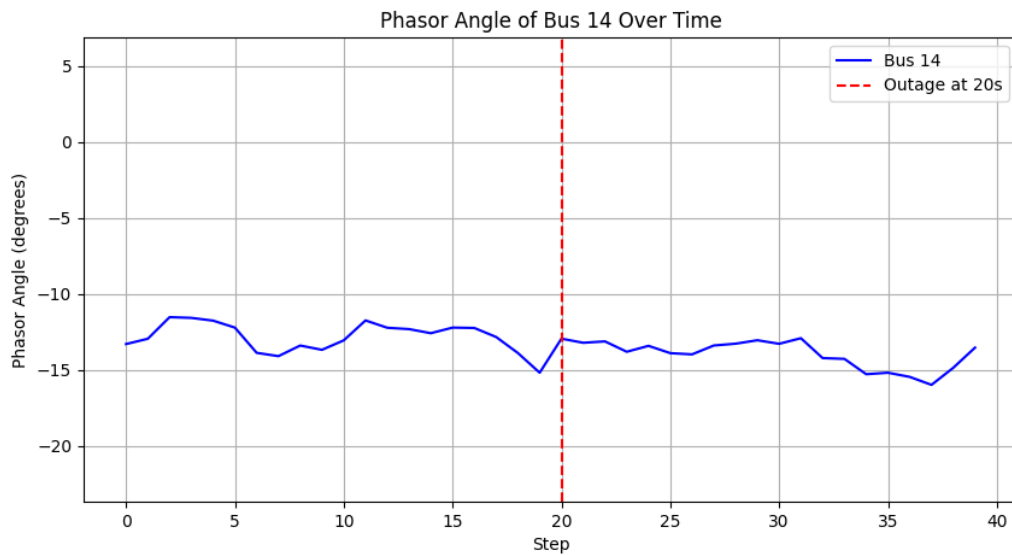


Figure 4: IEEE39 case where (2,17) and (14,15) are tripped at step 20

Training Process

The training pipeline was designed to evaluate both spatial and spectral GCN architectures followed by two FC layers (one extra over the original architecture), with a focus on finding the best-performing configuration for outage detection. To ensure fair comparison and robust tuning, the dataset was split into 70% training, 10% validation, 10% calibration, and 10% test sets, with a fixed random seed for reproducibility.

Hyperparameter Optimization

A large-scale hyperparameter search was conducted using **Optuna**, with 1000 trials exploring a broad design space:

- **Model type:** GFT vs. DFT-based architectures
- **Polynomial order:** 0, 1, 2
- **Loss functions:** Binary cross-entropy (BCE), Jeffrey's divergence, and DKL
- **Optimizers:** Adam and SGD
- **Schedulers:** StepLR, Cosine, ReduceLROnPlateau
- **Learning rate, weight decay, Batch size and dropout**

This automated search enabled systematic fine-tuning of architectural and training choices, balancing accuracy, stability, and efficiency.

Final Model Choice

The best-performing configuration was found to be the **GFT model with polynomial order 1** (similar to He and Cheng findings), trained with the Adam optimizer, DKL loss and a cosine scheduler.

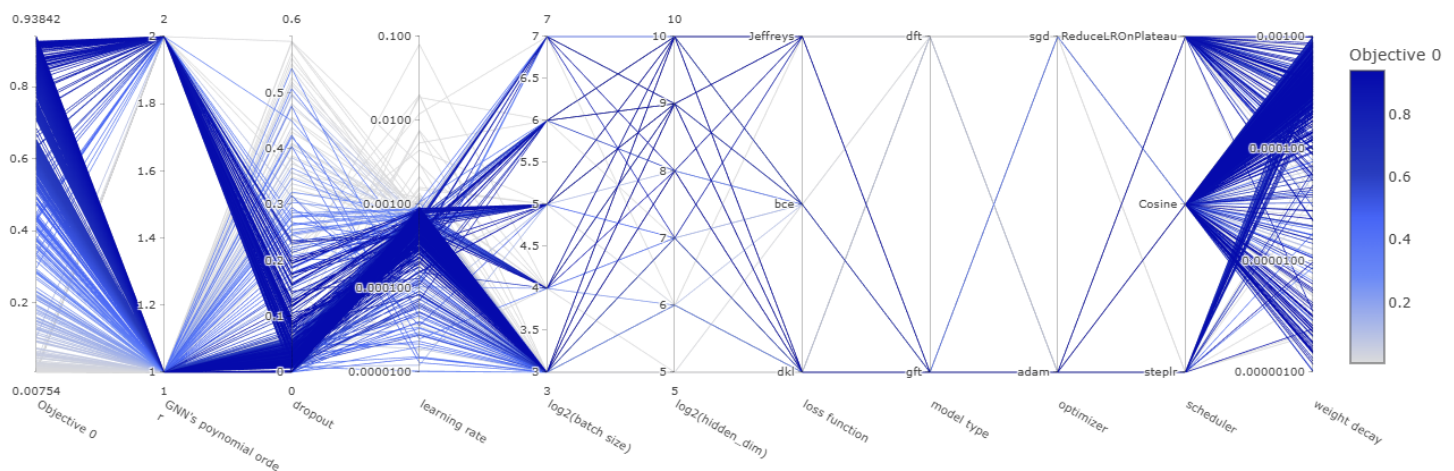


Figure 5: Optuna trials. Each line represents a choice of hyper-parameters (a single trial) where its color represents the validation accuracy

Conformal Prediction

Conformal prediction is a wrapper that adds a dependable measure of uncertainty to any machine learning model for an acceptable error rate α (e.g., 10 %). For every new input, the method returns a *set of plausible answers*, a class list (for classification) or a numeric interval (for regression), constructed so that the true class (true label) will fall inside that set at least $(1 - \alpha)$ of the time, assuming only that the data points are drawn independently. [3]



Figure 6: Illustration of RAPS conformal prediction on an image classification task. The size of the prediction set increases with task difficulty, reflecting the model's uncertainty in recognizing the true label

Conformity Scores

The key ingredient in CP is the **conformity score**, a measure of how well a candidate label “fits” a given input relative to the calibration data (opposite for nonconformity scores). For classification, this is often derived from model-predicted probabilities. Intuitively, classes with higher conformity are more likely to be included in the prediction set.

The Conformal Guarantee

Let $\mathcal{C} = \{(x_i, y_i) : i = 1, 2, \dots, n\}$ denote the calibration set and $q_{1-\alpha}$ the $(1 - \alpha)$ -quantile of nonconformity scores on $\mathcal{C}$:

$$q = \inf \{q : |\{i : S(x_i, y_i) \leq q\}| \geq \lceil (n + 1)(1 - \alpha) \rceil\}$$

Then, for a new input x , the conformal prediction set is defined as:

$$\Gamma_\alpha(x) = \{y \in \mathcal{Y} : S(x, y) \leq q_{1-\alpha}\}$$

The **fundamental theorem of conformal prediction** ensures that:

$$\Pr(y_{test} \in \Gamma_\alpha(x_{test})) \leq 1 - \alpha$$

without assumptions on the model architecture, loss function, or data distribution (beyond exchangeability). [3, section D]

Regularized Adaptive Prediction Sets (RAPS)

In this project, we employ the **RAPS** method [4], which refines standard CP by incorporating two modifications:

1. **Adaptive scores:** instead of relying only on the top probability, RAPS considers cumulative probability mass across ordered classes.
2. **Regularization:** an additive term encourages the method to include fewer irrelevant classes, leading to **smaller prediction sets** while still maintaining valid coverage.

Formally, given predicted class probabilities $(p_1, p_2, \dots, p_L)$ sorted in decreasing order, the nonconformity score is defined as:

$$S(x, y) = \sum_{i=1}^{r(y)-1} p_i + \lambda(r(y) - 1)$$

where $r(y)$ is the rank of the true label, and $\lambda > 0$ is the regularization parameter. The score penalizes both cumulative mass assigned to incorrect labels and the position of the true class, balancing **coverage** with **set size**.

Integration in Our Framework

We apply RAPS on top of the trained GCN models for outage detection. The output is not a single predicted line outage, but a **calibrated set of candidate outages**. This provides operators with a controlled trade-off: by selecting α , they can guarantee a desired level of reliability while keeping the prediction sets as compact as possible.

Results

Our experiments demonstrate the effectiveness of spectral GCNs for outage detection. Among the architectures tested, the Graph Fourier Transform (GFT) model with polynomial order 1 achieved the best baseline performance, reaching **92.79% top-3 accuracy** on the IEEE 39-bus test system.

With the RAPS conformal prediction method we achieved performance of **95.81% coverage**, while keeping the average prediction set compact at **3.42 candidate outages** on average. In practical terms, this means that the model outputs compact, calibrated candidate sets, ensuring operators can focus on a small number of highly probable contingencies.

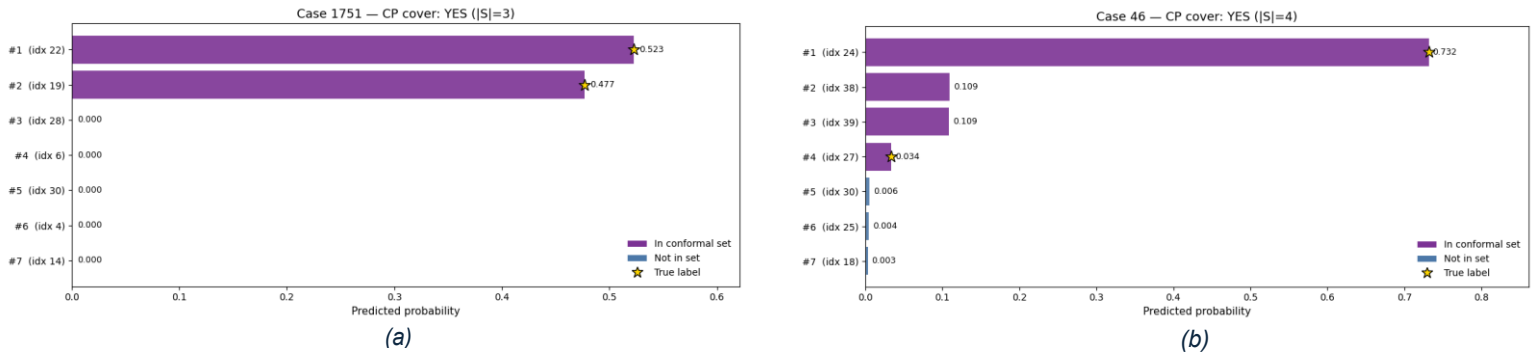


Figure 7: Results on IEEE39 cases. In (b) we see that line 27 that is tripped in this case is included in the prediction set but not included in the top3.

Conclusions and Future Work

This work demonstrates that graph convolutional networks, when combined with conformal prediction, offer an effective and scalable approach for detecting power line outages. Leveraging the graph-based characteristics of the power grid enables the models to recognize rare outage scenarios while maintaining high accuracy. With conformal prediction, our method achieved **95.81% coverage** while providing statistical guarantees on the predictions. In addition, a single forward pass of the GCN has computational complexity $O(N^2)$, amounting to only ~ 0.7 milliseconds on the 39-bus system (on CPU). This efficiency makes the approach attractive for real-time applications, especially when compared to classical methods that scale as $O(N^{3+k})$.

Through conformal prediction, we further enhance the reliability of the model by providing statistical coverage guarantees, while still allowing traditional methods to guide the ultimate decision.

While these results highlight the promise of GCNs for outage detection, several directions remain open for future exploration:

- **Learnable graph structure:** Instead of fixing the adjacency matrix from physical grid parameters and only learning polynomial filter weights, future work could allow the nonzero entries of the adjacency (edge weights) matrix to be learned directly. This would let the model adapt connectivity strengths based on data, potentially capturing effects such as fluctuating line impedances, varying load conditions, or hidden correlations not represented in the static grid model.
- **More realistic data:** Incorporate noisy or real PMU measurements, as well as outage events occurring at random times, to better reflect operational conditions.
- **Scalability:** Extend experiments to larger IEEE test systems and real-world grids to validate robustness at scale.
- **Hybrid architecture:** Explore combinations of spatial and spectral GCN approaches to leverage the strengths of both domains.
- **Additional measurement features:** Beyond phasor angle data, future models could incorporate other forms of PMU measurements. For example, including absolute voltage magnitudes as node features may provide complementary information on grid state and improve detection of outage events that are not fully captured by angle dynamics.

References

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- [5] E. Drayer and T. Routtenberg, "[Detection of False Data Injection Attacks in Smart Grids Based on Graph Signal Processing](#)," *IEEE Systems Journal*, vol. 14, no. 2, pp. 1886–1896, Jun. 2020.